FORMULA DOCUMENTATION — PRACTICAL PART

1. MEAN FORMULA

Definition

Mean is the arithmetic average of all observations in the dataset.

Formula:

$$\text{Mean} = \frac{\sum x}{n}$$

Where:

- Σx = sum of all values
- n = total number of observations

Purpose:
Used to calculate the average household income and average age.

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2. MEDIAN FORMULA

Definition

Median is the middle value when data is arranged in ascending or descending order.

Formula:

For odd number of observations:

$$\text{Median} = \left(\frac{n+1}{2}\right)^{th} value$$

For even number of observations:

$$\text{Median} = \frac{\text{Middle Value 1 + Middle Value 2}}{2}$$

Purpose:
Used to identify the center of the income distribution.

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3. MODE FORMULA

Definition

Mode is the value that occurs most frequently in the dataset.

Purpose:
Used to determine the most common household income or age group.

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4. RANGE FORMULA

Definition

Range measures the spread between highest and lowest values.

Formula:

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

Purpose:
Used to measure spread of household income.

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5. VARIANCE FORMULA

Definition

Variance measures how far data values are spread from the mean.

Formula:

$$\sigma^2 = \frac{\sum(x-\mu)^2}{N}$$

Where:

- x = data values
- μ = mean
- N = total observations

Purpose:
Used to understand variability in household income.

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6. STANDARD DEVIATION FORMULA

Definition

Standard deviation measures average dispersion of data from the mean.

Formula:

$$\sigma = \sqrt{\sigma^2}$$

Purpose:
Used to determine consistency in income values.

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7. INTERQUARTILE RANGE (IQR)

Definition

Interquartile Range measures spread of middle 50% of data.

Formula:

$$IQR = Q_3 - Q_1$$

Where:

- Q1 = First Quartile
- Q3 = Third Quartile

Purpose:
Used to analyze income spread and detect outliers.

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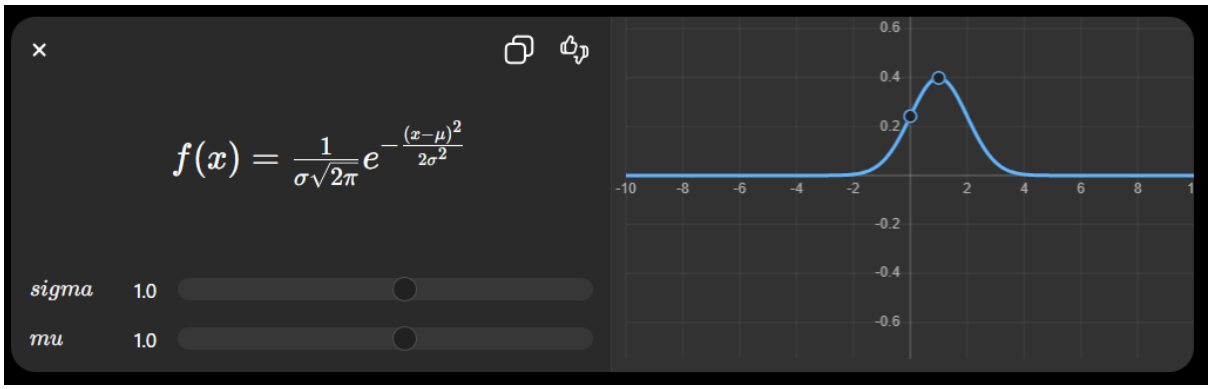
8. GAUSSIAN NORMAL DISTRIBUTION FORMULA

Definition

Gaussian distribution is a symmetric bell-shaped distribution.

Formula:



Where:

- μ = mean
- σ = standard deviation

Purpose:
Used to analyze whether income data follows normal distribution.

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9. SKEWNESS FORMULA

Definition

Skewness measures asymmetry of distribution.

Formula:

$$\text{Skewness} = \frac{\sum(x-\mu)^3}{N\sigma^3}$$

Purpose:
Used to determine whether income distribution is positively or negatively skewed.

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10. KURTOSIS FORMULA

Definition

Kurtosis measures peakedness or flatness of data distribution.

Formula:

$$\text{Kurtosis} = \frac{\sum(x-\mu)^4}{N\sigma^4}$$

Purpose:
Used to identify presence of outliers and distribution shape.

GRAPH DOCUMENTATION

1. HISTOGRAM

Definition

A histogram is a graph used to display distribution of numerical data using bars.

Purpose:

- Analyze income distribution
- Understand frequency of income ranges
- Detect skewness

Observation:
The histogram shows how household income values are distributed.

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2. KDE PLOT (Kernel Density Estimation)

Definition

KDE plot is a smooth curve representing probability density of data.

Purpose:

- Visualize data distribution smoothly
- Compare shape of income distribution

Observation:
The KDE curve helps understand concentration of income values.

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3. BOXPLOT

Definition

A boxplot visually represents spread and summary of data.

It displays:

- Minimum value
- Q1
- Median
- Q3
- Maximum value
- Outliers

Purpose:
Used to compare family size by education level.

Observation:
Boxplot helps identify spread and unusual observations.

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4. DISTRIBUTION CURVE FOR AGE VS INCOME

Definition

A distribution curve shows relationship between age and household income.

Purpose:

- Analyze trends between age and income
- Identify patterns in data

Observation:
The curve helps understand how income changes with age.

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5. GAUSSIAN NORMAL DISTRIBUTION CURVE

Definition

A Gaussian curve is a bell-shaped curve representing normal distribution.

Characteristics:

- Symmetric shape
- Mean = Median = Mode

Purpose:
Used to compare actual income distribution with normal distribution.

Observation:
Helps determine whether data follows normal distribution pattern.